

Gitlab with LDAP in samba AD

Integrating GitLab with a Samba Active Directory (AD) environment allows users to authenticate using their domain credentials.

LDAP will be installed along with samba AD. Install GitLab and change configuration.

Follow the steps:

Step-1: Check the LDAP.

```
# ldapsearch -x -H ldap://10.10.40.88 -b "dc=smb,dc=com"
```

```
siddharrth@master:~$ ldapsearch -x -H ldap://10.10.40.88 -b "dc=smb,dc=com"
# extended LDIF
#
# LDAPv3
# base <dc=smb,dc=com> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# search result
search: 2
result: 1 Operations error
text: 00002020: Operation unavailable without authentication
# numResponses: 1
```

```
siddharrth@master:~$ ldapsearch -H ldaps://10.10.40.88 -D "CN=Administrator,CN=Users,DC=smb,DC=com" -W -b "DC=smb,DC=com" "(objectClass=user)" dn -o tls_reqcert=never
Enter LDAP Password:
# extended LDIF
#
# LDAPv3
# base <DC=smb,DC=com> with scope subtree
# filter: (objectClass=user)
# requesting: dn
#
# Guest, Users, smb.com
dn: CN=Guest,CN=Users,DC=smb,DC=com
# Siddhu, Users, smb.com
dn: CN=Siddhu,CN=Users,DC=smb,DC=com
# user1, Users, smb.com
dn: CN=user1,CN=Users,DC=smb,DC=com
```

```
# search reference
ref: ldaps://smb.com/CN=Configuration,DC=smb,DC=com

# search reference
ref: ldaps://smb.com/DC=DomainDnsZones,DC=smb,DC=com

# search reference
ref: ldaps://smb.com/DC=ForestDnsZones,DC=smb,DC=com

# search result
search: 2
result: 0 Success

# numResponses: 14
# numEntries: 10
# numReferences: 3
```

Step-2: Install SSL certificates bundle, Timezone configuration for gitlab dependencies

```
# sudo apt update
```

```
# sudo apt install -y curl ca-certificates tzdata perl
```

```
siddharth@master:~$ sudo apt update
sudo apt install -y curl ca-certificates tzdata perl
[sudo] password for siddharth:
Hit:1 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.8).
curl set to manually installed.
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
tzdata is already the newest version (2026a-0ubuntu0.24.04.1).
tzdata set to manually installed.
perl is already the newest version (5.38.2-3.2ubuntu0.2).
perl set to manually installed.
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme at-spi2-common at-spi2-core dconf-gsettings-backend dconf-service fontconfig
  gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme javascript-common
  libatk-bridge2.0-0t64 libatk1.0-0t64 libatspi2.0-0t64 libcairo-gobject2 libcairo2 libcolord2 libdatrie1 libdconf1
  libepoxy0 libexpat1-dev libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgraphite2-3
  libgtk-3-0t64 libgtk-3-bin libgtk-3-common libharfbuzz0b libjs-jquery libjs-sphinxdoc libjs-underscore liblcms2-2
  libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpixman-1-0 libpython3-dev libpython3.12-dev librsvg2-2
  librsvg2-common libthai-data libthai0 libwayland-client0 libwayland-cursor0 libwayland-egl1 libxcb-render0
  libxcb-shm0 libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxi6 libxinerama1 libxrandr2 libxrender1 libxtst6
  python3-dev python3-wheel python3.12-dev session-migration tzdata-legacy ubuntu-mono x11-common zlib1g-dev
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Step-3: Download the Gitlab repo

```

sidharth@master:~$ curl https://packages.gitlab.com/install/repositories/gitlab/gitlab-ee/script.deb.sh | sudo bash
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           % Done    0     0      0      0      0     0      0      0
100 5291 100 5291    0     0  2247      0  0:00:02  0:00:02 --:--:-- 2248
Hit:1 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 https://packages.gitlab.com/gitlab/gitlab-ee/ubuntu/noble noble InRelease [16.9 kB]
Get:6 https://packages.gitlab.com/gitlab/gitlab-ee/ubuntu/noble noble/main amd64 Packages [14.1 kB]
Fetched 31.0 kB in 3s (12.1 kB/s)
Reading package lists... Done
Repository configured successfully.
Ready to install packages.
sidharth@master:~$

```

```
# sudo apt install gitlab-ee
```

```

Unpacking gitlab-ee (18.11.1-ee.0) ...
Setting up gitlab-ee (18.11.1-ee.0) ...
It looks like GitLab has not been configured yet; skipping the upgrade script.

      .
      :c:
      :ccc:
      :ccccc,
      .ccccccc
      cccccccc:
      cccccccc:
      :ccccccccc;.....'ccccccccc'
      cccccccccccccccccccccccccccccccccccc'
      :oo!ccccccccccccccccccccccccccccccccellooo
      oooooooooollcccccccccccccccccccccccccclooooooooo
      ;oooooooooollccccccccccccccccccccccloooooooooo!
      oooooooooooooollccccccccccccccloooooooooooooo.
      .ooooooooooooooooollccccccellooooooooooooo;
      cooooooooooooooooollloooooooooooooo.
      loooooooooodxxxxddoooooooooo.
      .ooodxxxxxxkxxxxxxdoo;
      .kkkkkkkkkkkkk:
      ;kkkkkkx
      :d

Thank you for installing GitLab!
GitLab was unable to detect a valid hostname for your instance.
Please configure a URL for your GitLab instance by setting 'external_url'
configuration in /etc/gitlab/gitlab.rb file.
Then, you can start your GitLab instance by running the following command:
sudo gitlab-ctl reconfigure

```

```
# sudo gitlab-ctl status
```

```
siddharrth@master:~$ sudo !!
sudo gitlab-ctl status
run: alertmanager: (pid 940) 1103s; run: log: (pid 926) 1103s
run: gitaly: (pid 939) 1103s; run: log: (pid 938) 1103s
run: gitlab-exporter: (pid 941) 1103s; run: log: (pid 923) 1103s
run: gitlab-kas: (pid 958) 1103s; run: log: (pid 947) 1103s
run: gitlab-workhorse: (pid 946) 1103s; run: log: (pid 928) 1103s
run: logrotate: (pid 954) 1103s; run: log: (pid 949) 1103s
run: nginx: (pid 942) 1103s; run: log: (pid 935) 1103s
run: node-exporter: (pid 945) 1103s; run: log: (pid 932) 1103s
run: postgres-exporter: (pid 952) 1103s; run: log: (pid 933) 1103s
run: postgresql: (pid 948) 1103s; run: log: (pid 930) 1103s
run: prometheus: (pid 943) 1103s; run: log: (pid 936) 1103s
run: puma: (pid 3322) 222s; run: log: (pid 950) 1103s
run: redis: (pid 951) 1103s; run: log: (pid 937) 1103s
run: redis-exporter: (pid 953) 1103s; run: log: (pid 934) 1103s
run: sidekiq: (pid 3027) 310s; run: log: (pid 927) 1103s
```

Step-6: Configure the Gitlab setup

```
# sudo vi /etc/gitlab/gitlab.rb
```

- Add this content

```
gitlab_rails['ldap_enabled'] = true

gitlab_rails['prevent_ldap_sign_in'] = false

gitlab_rails['ldap_servers'] = {

  'main' => {

    'label' => 'SambaAD',

    'host' => '10.10.40.88',

    'port' => 389,

    'uid' => 'sAMAccountName',

    'encryption' => 'start_tls',

    'verify_certificates' => false,

    'bind_dn' => 'CN=Administrator,CN=Users,DC=smb,DC=com',

    'password' => 'Optit@123',

    'base' => 'DC=smb,DC=com',

    'user_filter' => '(&(objectClass=user)!(objectClass=computer))',

    'attributes' => {

      'username' => ['sAMAccountName'],
```

```

'email' => ['mail', 'userPrincipalName'],

'name' => 'displayName',

'first_name' => 'givenName',

'last_name' => 'sn'

},

'lowercase_usernames' => true,

'active_directory' => true

}

```

EOS

```

gitlab_rails['ldap_enabled'] = true
gitlab_rails['prevent_ldap_sign_in'] = false

gitlab_rails['ldap_servers'] = {
  'main' => {
    'label' => 'SambaAD',
    'host' => '10.10.40.88',
    'port' => 389,
    'uid' => 'sAMAccountName',
    'encryption' => 'start_tls',
    'verify_certificates' => false,
    'bind_dn' => 'CN=Administrator,CN=Users,DC=smb,DC=com',
    'password' => 'Optit@123',
    'base' => 'DC=smb,DC=com',
    'user_filter' => '(&(objectClass=user)(!(objectClass=computer)))',
    'attributes' => {
      'username' => ['sAMAccountName'],
      'email' => ['mail', 'userPrincipalName'],
      'name' => 'displayName',
      'first_name' => 'givenName',
      'last_name' => 'sn'
    },
    'lowercase_usernames' => true,
    'active_directory' => true
  }
}

```

`"/etc/gitlab/gitlab.rb"` 3763L, 167354B

Step-7: Apply the gitlab configure or if changes made reconfigure

```
# sudo gitlab-ctl reconfigure
```

```
siddharth@master:~$ sudo gitlab-ctl reconfigure
[2026-04-30T04:54:10+00:00] INFO: Started Cinc Zero at chefzero://localhost:1 with repository at /opt/gitlab/embedded (One version per cookbook)
Cinc Client, version 18.3.0
Patents: https://www.chef.io/patents
Infra Phase starting
[2026-04-30T04:54:16+00:00] INFO: *** Cinc Client 18.3.0 ***
[2026-04-30T04:54:16+00:00] INFO: Platform: x86_64-linux
[2026-04-30T04:54:16+00:00] INFO: Cinc-client pid: 86226
[2026-04-30T04:54:18+00:00] INFO: Setting the run_list to ["recipe[gitlab-base]"] from CLI options
[2026-04-30T04:54:18+00:00] INFO: Run List is [recipe[gitlab-base]]
[2026-04-30T04:54:18+00:00] INFO: Run List expands to [gitlab-base]
[2026-04-30T04:54:18+00:00] INFO: Starting Cinc Client Run for master
[2026-04-30T04:54:18+00:00] INFO: Running start handlers
[2026-04-30T04:54:18+00:00] INFO: Start handlers complete.
Resolving cookbooks for run list: ["gitlab-base"]
[2026-04-30T04:54:20+00:00] INFO: Loading cookbooks [gitlab-base@0.1.0, package@0.1.0, gitlab@0.0.1, gitlab-ee@0.0.1, runit@5.1.7, logrotate@0.1.0, postgresql@0.1.0, redis@0.1.0, monitoring@0.1.0, registry@0.1.0, mattermost@0.1.0, consul@0.1.0, gitaly@0.1.0, praefect@0.1.0, gitlab-kas@0.1.0, gitlab-pages@0.1.0, letsencrypt@0.1.0, nginx@0.1.0, oak@0.1.0, patroni@0.1.0, pgbouncer@0.1.0, spamcheck@0.1.0, acme@4.2.2, crond@0.1.0]
Synchronizing cookbooks:
- gitlab-base (0.1.0)
- package (0.1.0)
- gitlab (0.0.1)
- gitlab-ee (0.0.1)
- runit (5.1.7)
- logrotate (0.1.0)
- postgresql (0.1.0)
```

```
- execute /opt/gitlab/bin/gitlab-rake cache:clear
[2026-04-28T07:44:14+00:00] INFO: Cinc Client Run complete in 102.617288109 seconds

Running handlers:
[2026-04-28T07:44:15+00:00] INFO: Running report handlers
Running handlers complete
[2026-04-28T07:44:15+00:00] INFO: Report handlers complete
Infra Phase complete, 6/931 resources updated in 01 minutes 49 seconds
gitlab Reconfigured!
```

- If, it is not working. Use this

ldapssearch -H ldaps://10.10.40.88 -D "CN=Administrator,CN=Users,DC=smb,DC=com" \

-W -b "DC=smb,DC=com" "(objectClass=user)" dn -o tls_reqcert=never

```
siddharth@master:~$ ldapssearch -H ldaps://10.10.40.88 \
-D "CN=Administrator,CN=Users,DC=smb,DC=com" \
-W \
-b "DC=smb,DC=com" \
"(objectClass=user)" dn \
-o tls_reqcert=never
```

```
Enter LDAP Password: Optit@123
# extended LDIF
#
# LDAPv3
# base <DC=smb,DC=com> with scope subtree
# filter: (objectClass=user)
# requesting: dn
#
# Guest, Users, smb.com
dn: CN=Guest,CN=Users,DC=smb,DC=com
# Siddhu, Users, smb.com
dn: CN=Siddhu,CN=Users,DC=smb,DC=com
```

Step-8: Check LDAP integration in GitLab

```
# sudo gitlab-rake gitlab:ldap:check
```

```
siddharrth@master:~$ sudo gitlab-rake gitlab:ldap:check

Checking LDAP ...

LDAP: ... Server: ldapmain
not verifying SSL hostname of LDAPS server '10.10.40.88:389'
LDAP authentication... Success

LDAP users with access to your GitLab server (only showing the first 100 results)
  DN: cn=guest,cn=users,dc=smb,dc=com      sAMAccountName: Guest
  DN: cn=siddhu,cn=users,dc=smb,dc=com     sAMAccountName: siddhu
  DN: cn=user1,cn=users,dc=smb,dc=com      sAMAccountName: user1
  DN: cn=administrator,cn=users,dc=smb,dc=com sAMAccountName: Administrator
  DN: cn=sid,cn=users,dc=smb,dc=com        sAMAccountName: sid
  DN: cn=ldapuser,cn=users,dc=smb,dc=com   sAMAccountName: ldapuser
  DN: cn=krbtgt,cn=users,dc=smb,dc=com     sAMAccountName: krbtgt

Checking LDAP ... Finished
```

Step-9: Check the web UI in browser

- Open any browser -> Enter the <gitlab-server-IP> ((i.e) 10.10.40.88)
- It will take some time to boot and start the depended package.
- Once, GitLab login appears. Select LDAP login option.
- Enter the credentials.

GitLab Enterprise Edition

SambaAD Standard

Username
Administrator

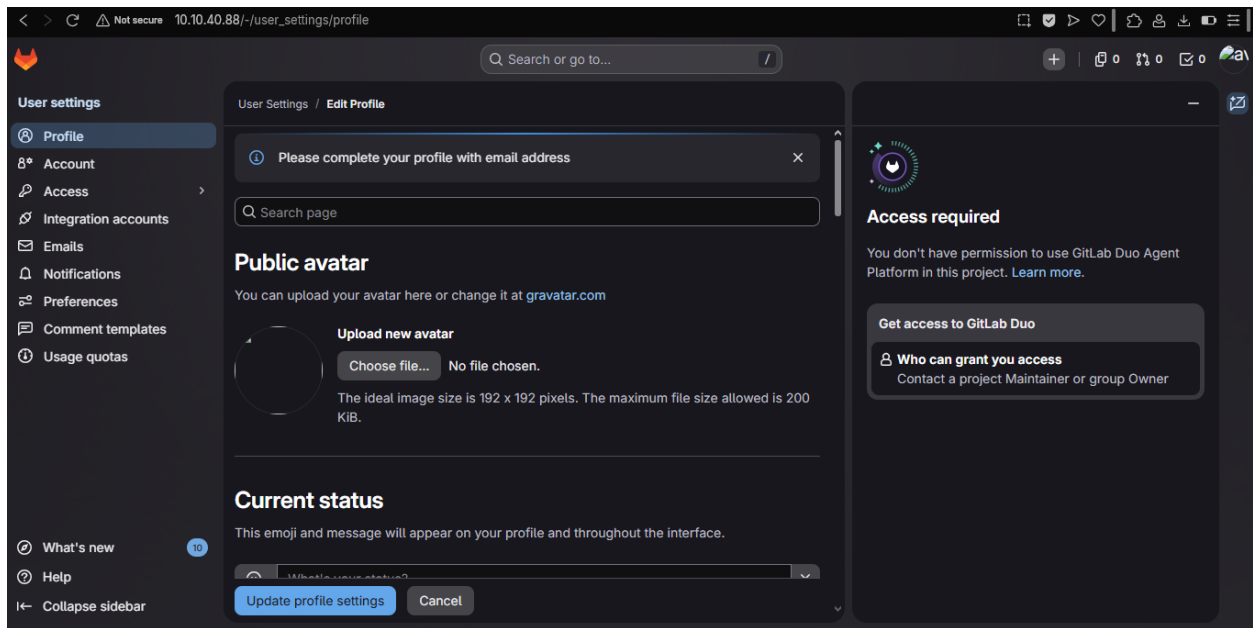
Password
.....

☐ Remember me

Sign in

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Troubleshoots:

- If, you face any issues use these commands to check everything is working fine

```
# sudo gitlab-ctl status puma
# sudo gitlab-ctl restart puma
# sudo gitlab-ctl restart sidekiq
# sudo netstat -tlnp | grep puma
# sudo lsof -i :8080
# sudo gitlab-ctl status
# sudo gitlab-ctl start
# sudo gitlab-ctl reconfigure
```